

Complete AWS Services Interview & Revision Guide

Amazon EC2

Basic Questions

What is EC2?

EC2 is a virtual server service in AWS used to run applications in the cloud.

What is an AMI?

AMI is a preconfigured template containing OS, applications, and settings used to launch EC2 instances.

What is an Instance Type?

Defines CPU, RAM, storage, and networking capacity of EC2.

What is a Key Pair?

Used for secure SSH login into EC2 instances.

What is Security Group?

A virtual firewall controlling inbound and outbound traffic for EC2 instances.

What is User Data?

A startup script automatically executed when EC2 launches.

What is Elastic IP?

A static public IP address for EC2.

What is Auto Scaling?

Automatically increases or decreases EC2 instances based on traffic.

What is Load Balancer?

Distributes traffic across multiple EC2 instances.

Advanced Questions

What is Spot Instance?

Unused AWS compute capacity available at low cost but can terminate anytime.

What is Reserved Instance?

Long-term EC2 reservation providing discounted pricing.

What is Dedicated Host?

Physical AWS server dedicated to one customer.

What is Placement Group?

Controls EC2 instance placement for better performance or fault tolerance.

What is EC2 Hibernate?

Stops instance while preserving RAM state.

Difference between Stop and Terminate?

Stop preserves instance; Terminate permanently deletes it.

What is ENI?

Elastic Network Interface providing networking for EC2.

What is EBS-Optimized Instance?

Provides dedicated bandwidth for EBS storage.

Scenario Questions

Scenario:

Website traffic suddenly increases. Use Auto Scaling with Load Balancer.

Scenario:

Need cheapest compute for temporary jobs. Use Spot Instances.

Scenario:

Need static IP for server. Attach Elastic IP.

Scenario:

Need highly available web application. Deploy EC2 across multiple Availability Zones.

Amazon S3

Basic Questions

What is S3?

S3 is object storage service for storing files and backups.

What is a Bucket?

Container used to store files in S3.

What is an Object?

Actual file stored in S3 bucket.

What is Versioning?

Stores multiple versions of objects.

What is Lifecycle Policy?

Automatically moves or deletes old objects.

What is Glacier?

Low-cost archival storage.

What is Static Website Hosting?

Hosting HTML/CSS/JS websites directly from S3.

Advanced Questions

What is Cross-Region Replication?

Automatically copies S3 objects to another AWS region.

What is S3 Transfer Acceleration?

Improves upload speed using edge locations.

What is Pre-Signed URL?

Temporary secure URL for private object access.

What is Server-Side Encryption?

AWS encrypts data before storing in S3.

What is Bucket Policy?

Access control policy attached to S3 bucket.

Scenario Questions

Scenario:

Need low-cost backup storage. Use Glacier.

Scenario:

Need global disaster recovery. Enable Cross-Region Replication.

Scenario:

Need public static website hosting. Use S3 Static Website Hosting.

IAM

Basic Questions

What is IAM?

IAM manages users, roles, and permissions securely in AWS.

What is IAM User?

Individual identity with AWS access.

What is IAM Group?

Collection of users with common permissions.

What is IAM Role?

Temporary permission identity assumed by services or users.

What is IAM Policy?

JSON document defining AWS permissions.

What is MFA?

Multi-Factor Authentication adding extra security.

Advanced Questions

Difference between IAM User and Role?

User is permanent identity; Role gives temporary permissions.

What is Assume Role?

Temporarily switching into another role.

What is Access Key?

Credential for programmatic AWS access.

What is Least Privilege Principle?

Giving only minimum required permissions.

Scenario Questions

Scenario:

Application needs S3 access securely. Attach IAM Role to EC2.

Scenario:

Developer needs AWS CLI access. Create IAM user with access keys.

Scenario:

Need extra account security. Enable MFA.

VPC

Basic Questions

What is VPC?

Private virtual network in AWS.

What is Subnet?

Smaller network inside VPC.

What is Public Subnet?

Subnet connected to internet.

What is Private Subnet?

Subnet without direct internet access.

What is Internet Gateway?

Allows internet connectivity for VPC.

What is Route Table?

Controls network routing.

What is NAT Gateway?

Allows private subnet outbound internet access.

Advanced Questions**What is VPC Peering?**

Private connection between two VPCs.

What is Transit Gateway?

Central hub connecting multiple VPCs.

What is NACL?

Subnet-level stateless firewall.

Difference between Security Group and NACL?

Security Group is stateful; NACL is stateless.

Scenario Questions**Scenario:**

Database should not be public. Place database in private subnet.

Scenario:

Connect two VPCs privately. Use VPC Peering.

Scenario:

Private server needs internet updates. Use NAT Gateway.

RDS

Basic Questions

What is RDS?

Managed relational database service.

Which databases does RDS support?

MySQL, PostgreSQL, Oracle, MariaDB, SQL Server.

What is Multi-AZ?

Standby database in another AZ for high availability.

What is Read Replica?

Read-only copy for scaling read traffic.

Advanced Questions

Difference between Multi-AZ and Read Replica?

Multi-AZ provides availability; Read Replica improves read scalability.

What are Automated Backups?

AWS automatically creates database backups.

What is Failover?

Automatic switch to standby database during failure.

Scenario Questions

Scenario:

Production database requires high availability. Enable Multi-AZ.

Scenario:

Application has heavy read traffic. Use Read Replica.

DynamoDB

Basic Questions

What is DynamoDB?

Managed NoSQL key-value database.

What is Partition Key?

Primary key used for data distribution.

What is On-Demand Capacity?

Automatically scales read/write capacity.

Advanced Questions

What is Global Secondary Index?

Additional query index for fast searches.

What is TTL?

Automatically deletes expired items.

Scenario Questions

Scenario:

Gaming app needs low-latency database. Use DynamoDB.

Scenario:

Need scalable serverless database. Use DynamoDB On-Demand.

CloudFront

Basic Questions

What is CloudFront?

AWS CDN service for faster global content delivery.

What is Edge Location?

AWS cache server near users.

Advanced Questions

What is Origin?

Backend source for CloudFront content.

What is Cache Invalidation?

Removing outdated cached content.

Scenario Questions

Scenario:

Global users experience slow website loading. Use CloudFront.

Scenario:

Updated website still shows old content. Create cache invalidation.

Route 53

Basic Questions

What is Route 53?

AWS DNS and domain management service.

What is Hosted Zone?

Container for DNS records.

What is A Record?

Maps domain to IP address.

What is CNAME?

Maps one domain to another domain.

Advanced Questions**What is Failover Routing?**

Automatically redirects traffic if endpoint fails.

What is Latency Routing?

Routes users to lowest latency region.

What is Weighted Routing?

Distributes traffic based on percentages.

Scenario Questions**Scenario:**

Need custom domain for website. Use Route 53.

Scenario:

Need disaster recovery DNS. Use Failover Routing.

CloudWatch**Basic Questions****What is CloudWatch?**

AWS monitoring and logging service.

What are Metrics?

Performance measurements like CPU utilization.

What is Alarm?

Triggers actions when thresholds exceed.

What are Logs?

Application and system log collection.

Advanced Questions

What is CloudWatch Dashboard?

Custom monitoring visualization dashboard.

What is EventBridge?

Event routing service for AWS events.

Scenario Questions

Scenario:

Need CPU alert notification. Create CloudWatch Alarm.

Scenario:

Need automated action on EC2 stop. Use EventBridge.

Lambda

Basic Questions

What is Lambda?

Serverless compute service.

What is Serverless?

No server management required.

What is Lambda Trigger?

Event that invokes Lambda function.

Advanced Questions

What is Cold Start?

Delay during Lambda initialization.

What is Lambda Timeout?

Maximum execution duration.

Scenario Questions

Scenario:

Run code when file uploads to S3. Use Lambda Trigger.

Scenario:

Need event-driven lightweight backend. Use Lambda.

EBS & EFS

Basic Questions

What is EBS?

Persistent block storage for EC2.

What is Snapshot?

Backup copy of EBS volume.

What is EFS?

Shared scalable file storage.

Advanced Questions

Difference between EBS and EFS?

EBS is single-instance block storage; EFS is shared file storage.

Difference between SSD and HDD?

SSD provides high performance; HDD provides higher throughput.

Scenario Questions

Scenario:

Need backup of EC2 storage. Create EBS Snapshot.

Scenario:

Multiple EC2 need same files. Use EFS.

ECS & EKS

Basic Questions

What is ECS?

AWS container orchestration service.

What is EKS?

Managed Kubernetes service.

What is Fargate?

Serverless container execution engine.

Advanced Questions

What is Task Definition?

Blueprint for running containers in ECS.

What is Kubernetes Control Plane?

Manages cluster operations.

Scenario Questions

Scenario:

Need managed Kubernetes cluster. Use EKS.

Scenario:

Run Docker containers without Kubernetes. Use ECS.

SNS & SQS

Basic Questions

What is SNS?

Notification service for alerts and messaging.

What is SQS?

Message queue service.

What is Queue?

Temporary message storage.

Advanced Questions

Difference between Standard and FIFO Queue?

FIFO preserves order; Standard provides higher throughput.

What is Topic in SNS?

Communication channel for notifications.

Scenario Questions

Scenario:

Send alert emails automatically. Use SNS.

Scenario:

Need decoupled application architecture. Use SQS.

Security Services

AWS Shield

Protection against DDoS attacks.

AWS WAF

Web Application Firewall filtering malicious traffic.

GuardDuty

Threat detection and continuous monitoring service.

Inspector

Security vulnerability assessment service.

KMS

Encryption key management service.

Scenario Questions

Scenario:

Need DDoS protection. Use AWS Shield.

Scenario:

Need malicious traffic filtering. Use AWS WAF.

Scenario:

Need threat detection. Use GuardDuty.

DevOps Services

CodePipeline

CI/CD automation service.

CodeBuild

Managed build service.

CodeDeploy

Automated deployment service.

Scenario Questions

Scenario:

Need automated deployment pipeline. Use CodePipeline.

Scenario:

Need automatic code deployment. Use CodeDeploy.

CloudFormation

What is CloudFormation?

Infrastructure as Code service.

Template Formats?

JSON and YAML.

Scenario:

Need repeatable infrastructure setup. Use CloudFormation.

CloudTrail

What is CloudTrail?

Tracks AWS API activity and user actions.

Scenario:

Need audit logs for investigation. Use CloudTrail.

AWS Organizations

What is AWS Organizations?

Central management for multiple AWS accounts.

Scenario:

Company manages many AWS accounts. Use AWS Organizations.

High Availability Architecture

Scenario:

Need scalable ecommerce website. Use:

- EC2
- Auto Scaling
- Load Balancer
- RDS Multi-AZ
- CloudFront
- Route 53
- WAF

Scenario:

Need serverless image processing. Use:

- S3
- Lambda
- API Gateway
- DynamoDB

Scenario:

Need secure multi-tier architecture. Use:

- VPC
- Public/Private Subnets
- NAT Gateway
- IAM Roles
- Security Groups

Shared Responsibility Model

AWS Responsibility

Security OF the cloud infrastructure.

Customer Responsibility

Security IN the cloud including IAM, OS, and data.

Regions & Availability Zones

What is Region?

Geographical AWS area.

What is Availability Zone?

Isolated datacenter inside a region.

Scenario:

Need high availability. Deploy resources across multiple Availability Zones.

Additional Advanced Interview Questions

EC2 Deep Dive

Difference between t2.micro and t3.micro?

T3 instances provide better burst performance and newer generation hardware.

What happens when EC2 reboots?

Temporary instance-store data may be lost; EBS data remains.

What is Instance Store?

Temporary storage attached physically to host machine.

Can we attach multiple EBS volumes?

Yes, multiple EBS volumes can attach to one EC2.

Scenario:

Need persistent storage after reboot. Use EBS volume.

S3 Deep Dive

Is S3 block storage?

No, S3 is object storage.

Maximum object size in S3?

5 TB.

What is Multipart Upload?

Large file upload in smaller chunks.

What is S3 Event Notification?

Triggers actions when object events occur.

Scenario:

Need automatic processing after upload. Use S3 Event Notification with Lambda.

IAM Deep Dive

What is Root User?

Main AWS account owner with full permissions.

Why avoid root account daily?

Root has unrestricted access and high security risk.

What is Inline Policy?

Policy directly attached to single user/group/role.

What is Managed Policy?

Reusable AWS-managed or customer-managed policy.

Scenario:

Need reusable permissions for many users. Use Managed Policies.

VPC Deep Dive

What is CIDR Block?

IP range assigned to VPC.

What is Bastion Host?

Secure EC2 used to access private servers.

Can private subnet access internet?

Yes, through NAT Gateway.

Scenario:

Need secure SSH access to private EC2. Use Bastion Host.

RDS Deep Dive

Can RDS scale automatically?

Storage can auto scale in supported engines.

What is Database Snapshot?

Manual backup of RDS database.

Can we SSH into RDS?

No, AWS manages OS access.

Scenario:

Need backup before upgrade. Create DB Snapshot.

DynamoDB Deep Dive

Difference between SQL and NoSQL?

SQL uses tables/relations; NoSQL provides flexible schema.

Does DynamoDB support auto scaling?

Yes.

Scenario:

Need massive scale with low latency. Use DynamoDB.

Lambda Deep Dive

Supported Lambda Languages?

Python, Node.js, Java, Go, .NET, Ruby.

Maximum Lambda Timeout?

15 minutes.

Scenario:

Need backend without managing servers. Use Lambda.

CloudFront Deep Dive

What content can CloudFront deliver?

Static and dynamic content.

Does CloudFront improve security?

Yes, using HTTPS and AWS Shield integration.

Scenario:

Need low latency global content delivery. Use CloudFront.

Route 53 Deep Dive

What is DNS?

System translating domain names into IP addresses.

What is TTL?

Time DNS records stay cached.

Scenario:

Need traffic split between two servers. Use Weighted Routing.

CloudWatch Deep Dive

Can CloudWatch monitor logs?

Yes.

What is Custom Metric?

User-defined monitoring metric.

Scenario:

Need notification when disk space low. Create CloudWatch Alarm.

CloudTrail Deep Dive

Does CloudTrail log API calls?

Yes, all AWS API activity can be logged.

Scenario:

Need to know who deleted resource. Use CloudTrail logs.

ECS / EKS Deep Dive

What is Pod in Kubernetes?

Smallest deployable Kubernetes unit.

What is Service in Kubernetes?

Exposes pods internally or externally.

Scenario:

Need container orchestration at scale. Use Kubernetes/EKS.

Security Deep Dive

What is Encryption at Rest?

Encrypting stored data.

What is Encryption in Transit?

Encrypting data while transferring.

What is Zero Trust?

Never trust; always verify access.

Scenario:

Need encrypted S3 bucket. Enable SSE-KMS.

DevOps Deep Dive

What is CI/CD?

Continuous Integration and Continuous Deployment.

What is Blue-Green Deployment?

Deployment strategy reducing downtime.

What is Rolling Deployment?

Gradually replacing old application versions.

Scenario:

Need zero downtime deployment. Use Blue-Green Deployment.

Cost Optimization Questions

How to reduce AWS cost?

Use Auto Scaling, Reserved Instances, and delete unused resources.

Cheapest EC2 option?

Spot Instances.

Cheapest long-term storage?

S3 Glacier.

Scenario:

Unused EBS volumes increasing bill. Delete unattached volumes.

Networking Advanced Questions

Difference between Public IP and Elastic IP?

Public IP changes; Elastic IP remains static.

What is Hybrid Cloud?

Combination of on-premises and cloud.

What is VPN Tunnel?

Encrypted connection between networks.

Scenario:

Need secure office-to-AWS connection. Use Site-to-Site VPN.

API Gateway Questions

What is API Gateway?

Managed service for creating and managing APIs.

Why use API Gateway?

Provides authentication, throttling, and monitoring.

Scenario:

Expose Lambda function as REST API. Use API Gateway.

Elastic Beanstalk Questions

What is Elastic Beanstalk?

Platform-as-a-Service for quick deployments.

Which languages supported?

Java, Python, Node.js, PHP, Go, .NET.

Scenario:

Need easy application deployment. Use Elastic Beanstalk.

AWS Backup Questions

What is AWS Backup?

Centralized backup management service.

Scenario:

Need automated backups across AWS services. Use AWS Backup.

Global Infrastructure Questions

What is Edge Location?

AWS location caching content near users.

What is Local Zone?

AWS infrastructure closer to metro areas.

Scenario:

Need ultra-low latency gaming. Use Local Zones.

How to Answer Scenario Questions

Scenario Question Meaning

Scenario questions test how you solve real-world AWS problems.

Interviewer gives:

- problem
- requirement
- business need

You explain:

- which AWS services to use
 - why to use them
 - architecture flow
-

Example Scenario Format

Question:

Your ecommerce website crashes during heavy traffic sales. How will you solve this?

Answer:

I will use:

- EC2 instances for application hosting
- Auto Scaling for automatic scaling

- Load Balancer for traffic distribution
- RDS Multi-AZ for database availability
- CloudFront for faster content delivery

This architecture improves scalability and high availability.

Easy Scenario Questions

Question:

You need to host a static portfolio website cheaply.

Answer:

I will use Amazon S3 static website hosting because it is low-cost, scalable, and easy to manage.

Question:

Users from different countries experience slow website loading.

Answer:

I will use CloudFront CDN because it caches content globally using edge locations.

Question:

You need secure login access for employees.

Answer:

I will create IAM users with limited permissions and enable MFA for additional security.

Question:

Your database should not be accessible publicly.

Answer:

I will place RDS inside private subnet and control access using Security Groups.

Question:

Application traffic increases automatically during daytime.

Answer:

I will configure Auto Scaling to automatically add EC2 instances based on CPU utilization.

Question:

Need automatic backup for important files.

Answer:

I will use S3 lifecycle policies and AWS Backup for automated backups.

Question:

Need serverless image processing system.

Answer:

I will use S3 for uploads and Lambda for automatic image processing.

Question:

Need low-cost long-term storage.

Answer:

I will use S3 Glacier because it provides low-cost archival storage.

Question:

Need to monitor CPU usage of EC2.

Answer:

I will use CloudWatch metrics and alarms for monitoring and notifications.

Question:

Need secure connection between office and AWS.

Answer:

I will use Site-to-Site VPN or AWS Direct Connect.

Question:

Need scalable Docker container deployment.

Answer:

I will use ECS or EKS depending on Kubernetes requirement.

Question:

Need to block malicious traffic.

Answer:

I will use AWS WAF and AWS Shield for application security.

Question:

Need audit logs for AWS account activities.

Answer:

I will use CloudTrail because it records AWS API activities.

Question:

Need decoupled architecture between services.

Answer:

I will use SQS queues for asynchronous communication.

Question:

Need email alerts when server fails.

Answer:

I will use CloudWatch alarms with SNS notifications.

Medium Level Scenario Questions**Question:**

Your application requires high availability and zero downtime.

Answer:

I will deploy EC2 instances across multiple Availability Zones with Load Balancer and Auto Scaling.

Question:

Need secure multi-tier architecture.

Answer:

I will use:

- Public subnet for Load Balancer
 - Private subnet for application servers
 - Private subnet for databases
 - NAT Gateway for internet access
 - Security Groups for protection
-

Question:

Application receives millions of requests.

Answer:

I will use Auto Scaling, Load Balancer, CloudFront, and DynamoDB for scalability.

Question:

Need disaster recovery architecture.

Answer:

I will enable Multi-AZ deployment, backups, and Cross-Region Replication.

Question:

Need automated CI/CD pipeline.

Answer:

I will use CodePipeline, CodeBuild, and CodeDeploy.

Question:

Need Kubernetes cluster in AWS.

Answer:

I will use Amazon EKS because it provides managed Kubernetes control plane.

Question:

Need event-driven architecture.

Answer:

I will use Lambda, SQS, SNS, and EventBridge.

Question:

Need secure encrypted storage.

Answer:

I will enable KMS encryption for S3, EBS, and RDS.

Question:

Need scalable REST APIs.

Answer:

I will use API Gateway with Lambda backend.

Advanced Scenario Questions

Question:

Design scalable ecommerce application architecture.

Answer:

I will use:

- Route 53 for DNS
 - CloudFront for CDN
 - Load Balancer for traffic distribution
 - Auto Scaling EC2 for scalability
 - RDS Multi-AZ for database
 - ElastiCache for caching
 - WAF and Shield for security
-

Question:

Design Netflix-like streaming platform.

Answer:

I will use:

- S3 for video storage
 - CloudFront for global streaming
 - EC2/ECS for backend
 - DynamoDB for metadata
 - Auto Scaling for scaling
-

Question:

Design secure banking application.

Answer:

I will use:

- IAM with least privilege
 - KMS encryption
 - WAF and Shield
 - CloudTrail logging
 - Multi-AZ architecture
 - Private subnets for databases
-

Question:

Design chat application architecture.

Answer:

I will use:

- API Gateway
 - Lambda
 - DynamoDB
 - WebSockets
 - SQS for asynchronous processing
-

Real Architecture Scenarios

Scenario:

Build Netflix-like video streaming platform. Use:

- S3
- CloudFront
- Auto Scaling
- Load Balancer
- DynamoDB

Scenario:

Build scalable chat application. Use:

- API Gateway
- Lambda
- DynamoDB
- SQS

Scenario:

Build secure banking application. Use:

- WAF
 - Shield
 - KMS
 - CloudTrail
 - IAM
-

Rapid Fire Interview Questions

Difference between EBS and S3?

EBS is block storage; S3 is object storage.

Difference between Scaling Up and Scaling Out?

Scaling Up increases server power; Scaling Out adds servers.

Difference between Horizontal and Vertical Scaling?

Horizontal adds machines; Vertical upgrades machine size.

Difference between Public and Private Subnet?

Public subnet has internet access; Private subnet does not.

Difference between ECS and EKS?

ECS is AWS container service; EKS is Kubernetes service.

Difference between SNS and SQS?

SNS sends notifications; SQS stores messages.

Difference between CloudWatch and CloudTrail?

CloudWatch monitors resources; CloudTrail logs API actions.

Difference between Security Group and NACL?

Security Group is stateful; NACL is stateless.

Difference between Read Replica and Multi-AZ?

Read Replica scales reads; Multi-AZ provides failover.

Difference between Root Volume and Additional Volume?

Root volume contains OS; additional volume stores extra data.

FINAL INTERVIEW PRACTICE BANK (Basic → Advanced)

EC2 Practice Questions

Q1. What is EC2?

EC2 is a virtual server in AWS used to run applications in the cloud.

Q2. What is Auto Scaling?

Automatically increases or decreases EC2 instances based on demand.

Q3. What is Load Balancer?

It distributes traffic across multiple EC2 instances to avoid overload.

Q4. Scenario:

Website traffic increases suddenly. Answer: Use Auto Scaling with Load Balancer to handle scaling and distribute traffic efficiently.

Q5. What is AMI?

Preconfigured template used to launch EC2 instances.

Q6. What is Security Group?

Acts as a virtual firewall controlling inbound and outbound traffic.

Q7. 20-30 Word Answer:

Auto Scaling increases EC2 instances automatically and Load Balancer distributes traffic evenly to ensure high availability and performance during traffic spikes.

S3 Practice Questions

Q1. What is S3?

Object storage service used to store files in AWS.

Q2. What is S3 Bucket?

A container that stores objects/files.

Q3. Scenario:

Store images for website. Answer: Use S3 bucket because it is scalable, durable, and cost-effective.

Q4. What is Glacier?

Low-cost archival storage for long-term data.

Q5. 20-30 Word Answer:

S3 is highly durable object storage used for storing static files, backups, and media with high availability and scalability.

IAM Practice Questions**Q1. What is IAM?**

Service to manage users, roles, and permissions in AWS.

Q2. What is IAM Role?

Temporary permission identity used by AWS services.

Q3. Scenario:

Give EC2 access to S3 securely. Answer: Use IAM Role attached to EC2 instance.

Q4. What is MFA?

Extra security layer using OTP verification.

Q5. 20-30 Word Answer:

IAM provides secure access control using users, roles, and policies ensuring least privilege access to AWS resources.

VPC Practice Questions**Q1. What is VPC?**

Virtual Private Cloud providing isolated network in AWS.

Q2. What is Subnet?

Smaller network inside VPC.

Q3. Scenario:

Database should not be public. Answer: Place database in private subnet for security.

Q4. What is NAT Gateway?

Allows private subnet to access internet securely.

Q5. 20-30 Word Answer:

VPC provides isolated networking environment with public and private subnets for secure and controlled AWS resource access.

RDS Practice Questions**Q1. What is RDS?**

Managed relational database service.

Q2. What is Multi-AZ?

Provides high availability with standby database.

Q3. Scenario:

Need highly available database. Answer: Use RDS Multi-AZ deployment.

Q4. What is Read Replica?

Used to scale read operations.

Q5. 20-30 Word Answer:

RDS is a managed database service providing automated backups, scaling, and high availability with Multi-AZ support.

Lambda Practice Questions**Q1. What is Lambda?**

Serverless compute service running code without servers.

Q2. Scenario:

Run code when file uploaded. Answer: Use Lambda triggered by S3 event.

Q3. What is Serverless?

No server management required by user.

Q4. 20-30 Word Answer:

AWS Lambda runs code on demand without managing servers and scales automatically based on events.

CloudFront Practice Questions

Q1. What is CloudFront?

Content Delivery Network for fast content delivery.

Q2. Scenario:

Website slow globally. Answer: Use CloudFront CDN to cache content near users.

Q3. 20-30 Word Answer:

CloudFront improves website performance by caching content at edge locations globally.

Route 53 Practice Questions

Q1. What is Route 53?

AWS DNS service.

Q2. Scenario:

Connect domain to website. Answer: Use Route 53 DNS records.

Q3. 20-30 Word Answer:

Route 53 is a scalable DNS service used to route traffic to AWS resources efficiently.

CloudWatch Practice Questions

Q1. What is CloudWatch?

Monitoring and logging service in AWS.

Q2. Scenario:

CPU usage high alert needed. Answer: Use CloudWatch Alarm.

Q3. 20-30 Word Answer:

CloudWatch monitors AWS resources and triggers alerts based on metrics and logs.

Security Practice Questions**Q1. What is AWS Shield?**

Protection against DDoS attacks.

Q2. What is WAF?

Web Application Firewall for filtering traffic.

Q3. Scenario:

Block malicious traffic. Answer: Use AWS WAF and Shield.

Q4. 20-30 Word Answer:

AWS provides security services like WAF, Shield, and KMS to protect applications and data.

FINAL RAPID SCENARIO PRACTICE**Q1. Build scalable website**

Answer: EC2 + Auto Scaling + Load Balancer + RDS + CloudFront

Q2. Build serverless app

Answer: S3 + Lambda + API Gateway + DynamoDB

Q3. Secure architecture

Answer: IAM + KMS + VPC + Private Subnets + WAF

Q4. High availability system

Answer: Multi-AZ + Load Balancer + Auto Scaling

FINAL 20-30 WORD MASTER ANSWER TEMPLATE

"In this scenario, I will use AWS services like EC2, Load Balancer, and Auto Scaling to ensure scalability, high availability, and fault tolerance based on system requirements."

IMPORTANT: SCENARIO ANSWERS LOCATION (CLARIFICATION)

Scenario answers are already included in this document in multiple sections:

1. "How to Answer Scenario Questions" section

- Explains how to structure answers
- Gives sample format and example

2. Each Service Section (EC2, S3, IAM, VPC, etc.)

- Under each service you will find:
- Scenario Question
- Direct Answer (which AWS service to use and why)

3. FINAL PRACTICE BANK

- Contains mixed scenario + direct answers for revision

👉 So you DO NOT need to search separately — scenarios are already embedded with every service.

EXTRA ADVANCED + INTERVIEW MASTER QUESTIONS (NEW)

EC2 Advanced Master Questions

Q. What is ENI used for in EC2?

ENI provides multiple private IPs and network flexibility for EC2 instances.

Q. What is EC2 Metadata?

Information about instance like IP, AMI, instance type accessible inside instance.

Q. What happens if EC2 host fails?

Instance may be lost; EBS backed instances can be recovered.

Q. Scenario:

Need high network performance EC2. Answer: Use enhanced networking and placement groups.

S3 Advanced Master Questions

Q. What is S3 consistency model?

S3 provides strong read-after-write consistency for all objects.

Q. What is S3 Object Lock?

Prevents object deletion for compliance requirements.

Q. Scenario:

Need to prevent file deletion for 5 years. Answer: Use S3 Object Lock with retention policy.

IAM Advanced Master Questions

Q. What is IAM Permission Boundary?

Limits maximum permissions a user/role can get.

Q. What is STS (Security Token Service)?

Provides temporary credentials for AWS access.

Q. Scenario:

External partner needs temporary access. Answer: Use STS AssumeRole.

VPC Advanced Master Questions

Q. What is VPC Flow Logs?

Captures network traffic information.

Q. What is Transit VPC?

Central VPC connecting multiple networks.

Q. Scenario:

Need to monitor network traffic. Answer: Enable VPC Flow Logs.

RDS Advanced Master Questions

Q. What is RDS Encryption?

Encrypts database using AWS KMS.

Q. What is DB Parameter Group?

Controls database engine configuration.

Q. Scenario:

Need encrypted database. Answer: Enable RDS encryption with KMS.

Lambda Advanced Master Questions

Q. What is Lambda concurrency?

Number of executions running at same time.

Q. What is Provisioned Concurrency?

Pre-warmed Lambda for low latency.

Q. Scenario:

Need low latency API response. Answer: Use Provisioned Concurrency.

CloudFront Advanced Master Questions

Q. What is Origin Shield?

Extra caching layer between edge and origin.

Q. What is Geo Restriction?

Block or allow countries.

Q. Scenario:

Block traffic from specific countries. Answer: Use CloudFront Geo Restriction.

Route 53 Advanced Master Questions

Q. What is Health Check in Route 53?

Monitors endpoint health.

Q. Scenario:

Redirect traffic if server fails. Answer: Use Route 53 Failover Routing with health check.

CloudWatch Advanced Master Questions

Q. What is Log Insights?

Query and analyze logs.

Q. Scenario:

Find error logs quickly. Answer: Use CloudWatch Logs Insights.

Security Advanced Master Questions

Q. What is KMS Key Rotation?

Automatically changing encryption keys.

Q. Scenario:

Need secure encryption lifecycle. Answer: Use KMS with automatic key rotation.

FINAL SUPER MASTER INTERVIEW QUESTIONS (MIX)

Q. Design highly scalable system for 10M users

Answer: Use CloudFront + ALB + Auto Scaling + RDS Multi-AZ + ElastiCache + Route 53

Q. Design real-time chat system

Answer: Use API Gateway WebSocket + Lambda + DynamoDB + SQS

Q. Design secure enterprise system

Answer: Use IAM + KMS + WAF + CloudTrail + VPC + Private Subnets

Q. Design cost optimized system

Answer: Use Spot Instances + S3 Glacier + Auto Scaling + Serverless architecture

FINAL NOTE

This document now contains:

- Basic Questions
- Advanced Questions
- Scenario Questions
- 20–30 word answers
- Deep architecture questions
- Real interview patterns

👉 You can now directly revise and practice for interviews from this single file.